DAMS SECTOR DESIGN BASIS FRAMEWORK

Design Basis, Design Basis Accident, and Military-Action Extension Methodology
Function-Primary Facility-Class Architecture — the Stored-Energy Sector

DBA-DAMS: Dams Sector Parent Document

A standalone sector series in the Design Basis for AI Governance in Critical Infrastructure

Version 0.4 — DRAFT

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

Section 1 — Purpose, Scope, and Series Architecture

1.1 Purpose

This document establishes the Design Basis Framework for the National Infrastructure Protection Plan (NIPP) Dams sector. It is the parent instrument for the DBA-DAMS series and applies the DB → DBA → DBA-MA three-layer analytical progression to the water-retention and control structures whose failure releases stored energy on a catastrophic scale: dams, reservoirs, and the levees and other retention works that protect populations from water.

1.2 What This Document Does Not Do

This document does not supersede dam-safety regulation or the Emergency Action Plan requirements. It does not restate the Bright Line or Command Broker, inherited by reference from FD-BL. It does not re-scope the multi-purpose functions a dam serves: hydroelectric generation is governed by the electric series (DBA-ES-HY-FC1), water supply and storage by the water series (DBA-W-FC3), and navigation locks by the transportation series, each inherited here as a boundary. DBA-DAMS governs the structure and the dam-safety consequence those functions share.

1.3 Function-Primary Organizing Principle

Placement follows the consequence chain. The Dams sector is organized by function — the retention structure and its water-control works, the reservoir, and the broader levee and flood-control scope — and each is placed by the downstream consequence its failure produces, sub-indexed by hazard-potential classification.

1.4 Series Placement — DBA-DAMS as a Standalone Sector Series

DBA-DAMS is a standalone sector series and the sixteenth NIPP sector represented in the corpus. It inherits the DB → DBA → DBA-MA methodology from the DBA-MA parent and the Bright Line and Command-Broker architecture from FD-BL, and it applies the mandatory Military-Action Extension Applicability Determination.

1.5 The Stored-Energy Sector — and the Consolidation of Dam Safety

The Dams sector carries a consequence unlike any other in the framework. A dam holds back an immense mass of water, and its purpose is to keep stored potential energy in reserve; its failure is the sudden, uncontrolled release of that energy as a catastrophic downstream flood. The consequence is not the loss of a service that recovers on a timeline — it is a peak, near-instantaneous, mass-casualty event whose severity is set by the population in the inundation zone. This is a stored-energy release consequence, and the sector taxonomy names it directly as the high consequence of failure that governs dam operation. The 2023 Kakhovka dam breach is its reference event: a dam failure used as an instrument of deliberate action, flooding a region

and, in a coupling the corpus has already noted, degrading the cooling-water source of a major nuclear plant. Kakhovka is to this sector what Zaporizhzhia is to the nuclear series.

The second feature of the sector is that a dam is inherently multi-purpose, and its purposes are already governed by other series. The same structure generates hydroelectric power (governed by DBA-ES-HY-FC1), stores water supply (governed by DBA-W-FC3), and supports navigation through its locks (governed by the transportation series). Each of those series touches the dam through the function it owns, but none of them owns the dam-safety consequence — the structural integrity and the catastrophic-release event that all of them depend on. DBA-DAMS is the sector that consolidates that consequence: it is the home of the structure and of the downstream-flood design basis, and it stands in a boundary relationship with the electric, water, and transportation series for the functions the structure serves.

Section 2 — The Dams Sector and Its Regulatory Context

2.1 A Multi-Owner, Dam-Safety-Governed Sector

The Dams sector is owned by a mix of federal agencies, states, utilities, and private parties, and is governed by an established dam-safety apparatus. The Federal Energy Regulatory Commission licenses and oversees non-federal hydroelectric projects; the Army Corps of Engineers is responsible for the safety of the dams it operates and maintains the national inventory; the Bureau of Reclamation oversees federal dams in the West; and state dam-safety programs regulate the large majority of dams. The Emergency Action Plan requirement — the downstream-warning and evacuation plan calibrated to the inundation zone — is the sector's signature safety instrument.

2.2 Hazard-Potential Classification as the Criticality Anchor

The sector carries a built-in criticality taxonomy: the hazard-potential classification that rates a dam high, significant, or low hazard-potential according to the downstream consequence of its failure, independent of the dam's condition. This is the dam-sector analog of the anchors the other frameworks use — the UN/DOT hazard class in chemical, the systemic designation in finance — an existing, authoritative, consequence-based classification to which DBA-DAMS anchors its design basis tiering rather than constructing a bespoke one.

2.3 The Regulatory Gap

The dam-safety apparatus is mature for accidental and hydrologic failure — for overtopping, seepage, and structural deterioration — but it was not built for the deliberate breach. The Emergency Action Plan assumes a failure that develops observably; it does not assume an adversary who selects the moment, the water level, and the coincident conditions to maximize the downstream consequence. The design basis this framework begins is the extension of dam safety to deliberate action, the gap the Kakhovka event exposed.

Section 3 — Function-Primary Taxonomy

3.1 The Function Classes and the Hazard-Potential Sub-Index

The sector is decomposed into three function classes — dam structures and their water-control works, reservoirs and appurtenant works, and the broader levee and retention scope — and one coordination class, the dam-safety-and-emergency-action layer. Each is sub-indexed by hazard-potential classification, so that the design basis load a given structure carries reflects the downstream population its failure would place at risk.

3.2 Dam Structures & Water-Control Works (DS)

The dam-structure function covers the retention structures themselves — concrete gravity, arch, and embankment dams — together with their spillways, gates and valves, and outlet works. Its governing consequence is the catastrophic release of the impounded water, and its design basis addresses both the structural integrity of the barrier and the controllability of the release: an adversary may attack the structure to breach it or the gates and controls to release it, and both paths lead to the downstream-flood consequence. It is the highest-consequence class of the sector.

3.3 Reservoirs & Appurtenant Works (RM)

The reservoir function covers the impounded storage itself, the intake and outlet structures, the level-control works, and the recreation infrastructure the reservoir carries. Its governing consequence is the loss of control over the stored mass — the reservoir level being the variable that sets the energy available for release — and it is the function through which the multi-purpose interfaces with the water and electric series are managed.

3.4 Levees, Floodwalls & Retention Works (LV)

The levee function covers the levees, floodwalls, and hurricane barriers that protect populations from water, and the tailings impoundments that retain industrial waste. Its governing consequence is the failure of protection — a levee breach that floods the area it was built to defend, or a tailings-impoundment failure that releases contained waste — and it extends the sector beyond the dam to the full water-retention-and-control scope the NIPP definition carries.

3.5 OpsCo — Dam Safety & Emergency Action (coordination class)

The coordination function is carried by a coordination class spanning the dam-safety programs, the Emergency Action Plan framework, and the SCADA, reservoir-level, and seismic monitoring systems that detect a developing failure. It is the sector's detection-and-warning layer — the function whose adequacy determines whether a downstream population is warned in time — and the locus at which the AI-Governance Design Basis is implemented: the Bright Line placement and Command-Broker obligation for the machine-learning and generative-AI systems in gate

control, reservoir optimization, and failure prediction, where an incorrect or compromised automated gate action is itself a release event.

† The DS / RM / LV and OpsCo designators were ratified 2026-08-09.

3.6 The Inundation Consequence Measure

The sector's consequence measure is not a duration but a peak. The catastrophic-release consequence is measured by the inundation zone and the population at risk within it — the same basis the hazard-potential classification uses — assessed for the deliberate case at the water level and coincident conditions an adversary would select. As with the other non-duration consequences the corpus has accumulated, the design basis states the peak consequence directly rather than a recovery timeline; recovery of the structure is a separate, and secondary, RT_HVLLC-style consideration. On the author's decision of 12 October 2026 the shared inundation consequence model that develops this measure for the deliberate case is held provisionally at *Dam Structures and Water-Control Works* (DBA-DAMS-DS-FC1) §4, until this framework confers a designation on a standalone model instrument. An instrument that needs the model cites that section.

Section 4 — The Dams Sector Design Basis

[Skeleton — to be developed.] The design envelope will be organized by event category, spanning the hydrologic, structural, and deliberate-breach cases, with a consequence-severity inclusion criterion consistent with the series.

4.1 Design Basis Principles

The criterion for inclusion is downstream-consequence severity — potential for a catastrophic release whose inundation zone places a population at risk of mass-casualty flooding — not event likelihood, consistent with the hazard-potential classification's own consequence-first logic. Candidate bounding events, to be developed: a deliberate breach of a high-hazard-potential dam; an uncontrolled gate or spillway release; a levee or tailings-impoundment failure; and the compound event that times a release to coincide with a downstream condition — a flood, or the loss of a dependent facility such as the nuclear cooling-water case the Kakhovka event demonstrated.

Section 5 — Dual Design Basis: Bright Line and Command Broker

[Skeleton — to be developed on the DBA-W §5.5 model.]

Dam operation is automating — SCADA-driven gate control, reservoir-optimization algorithms, and machine-learning failure prediction. The Bright Line must be placed for any action of a component within scope of FD-LD §4.3 whose incorrect performance could initiate or fail to prevent a release at or above the severity threshold — a gate-control command, a reservoir-drawdown decision, a spillway actuation — with the architecturally enforced authorization of a

qualified operator as Command Broker. Gate control is the sharpest case: an automated or compromised gate opening at a high reservoir level is a release event, and it must sit above the Bright Line without exception. Determinism decides scope under FD-LD §4.3; it does not decide placement.

Section 6 — Cross-Sector Dependencies and the Consequence Elevation Rule

6.1 Inward View — Power, Communications, and Weather

Dams depend on energy for station-service and backup generation power, on communications for the SCADA telemetry and the emergency-notification systems on which downstream warning depends, on transportation for maintenance access, and on weather services for the precipitation and snowmelt forecasting that governs reservoir management. The communications dependency is the most consequence-critical: the Emergency Action Plan's downstream warning fails if the notification path fails, so the communications-degradation baseline is a design basis condition for the warning function.

6.2 Outward View — the Downstream Population and the Multi-Purpose Dependents

The outward consequence of the sector runs first to the downstream population in the inundation zone — the mass-casualty flood — and second to the multi-purpose dependents: a dam failure simultaneously ends the hydroelectric generation the electric series counts, the water supply the water series stores, and the navigation the transportation series carries. A single structural failure is therefore a multi-sector event, and the boundary series each carry a dependency on the dam's integrity that this framework's consequence tiering elevates.

6.3 Named Boundaries

Three boundaries are named, and together they explain why this sector was previously only partially represented. Hydroelectric generation at a dam is DBA-ES-HY-FC1; water supply and storage is DBA-W-FC3; and navigation locks are the transportation series. Each already touched the dam through its own function, but the dam-safety consequence — the catastrophic release — belongs to none of them and is consolidated here. The boundary in each case runs at the structure: the served function is the partner series', and the integrity of the structure and the release event are DBA-DAMS.

Section 7 — Military-Action Extension Applicability

[Skeleton.] The Dams sector is the sector whose military-action case the corpus has already encountered empirically: the 2023 Kakhovka breach is the reference event, a dam failure used as an instrument of war to flood a region and degrade a dependent nuclear facility. The persistence and precision characteristics dominate — the deliberate selection of the moment and the water level to maximize downstream

consequence — and the sector-level presumption toward MA applicability is strong for all high-hazard-potential structures. Application remains a per-facility determination under the mandatory two-factor test — geolocation and strategic value — with strategic value driven by the inundation-zone population and by the multi-purpose dependents downstream.

Section 8 — Facility Class Register and Inheritance Model

8.1 Inheritance Model

Facility-class instruments of every type in the DBA-DAMS series inherit the threat taxonomy, bounding methodology, consequence-analysis structure, adequacy standards, and Bright Line / Command-Broker architecture established in this framework. Each instrument adds the structure-specific consequence chain, expressed in inundation-zone terms, and the boundary relationship with the electric, water, and transportation series for the functions the structure serves.

8.2 Facility Class Register

The following table is the current facility-class register for the DBA-DAMS series. DBA-DAMS-DS-FC1 has issued as a working draft and the other function classes are at Anticipated status; the hydroelectric, water-supply, and navigation functions are shown as governed by their partner series. Designators were ratified 2026-08-09 and sub-indexed by hazard-potential classification.

Designator	Function class (function-primary)	Governing characteristic	Status	Primary regulatory anchor
DBA-DAMS-DS-FC1	Dam structures & water-control works — concrete gravity, arch & embankment dams, spillways, gates & valves, outlet works	stored-energy catastrophic release; hazard-potential classified	Issued (working draft)	FERC; USACE; Bureau of Reclamation; state
DBA-DAMS-RM-FC2	Reservoirs & appurtenant works — impounded storage, intake & outlet structures, level control, recreation	stored mass; multi-purpose interfaces	Anticipated	FERC; Reclamation; state dam safety
DBA-DAMS-LV-FC3	Levees, floodwalls, hurricane barriers & tailings impoundments — the broader water-retention & control scope	flood-protection failure; containment release	Anticipated	USACE; state; MSHA (tailings)
DBA-DAMS-OpsCo†	Dam safety & emergency-action coordination (coordination class) — dam-safety programs, Emergency Action Plans, SCADA & seismic monitoring	downstream warning & consequence coordination	Anticipated (architecture)	USACE; FERC; state dam-safety programs
(boundaries)	Hydro generation → DBA-ES-HY-FC1 · water supply/storage → DBA-W-FC3 · navigation locks → DBA-TRAN	the dam serves functions owned by other series	Governed elsewhere	see those series

Designators were ratified 2026-08-09 at v0.2. Facility-class numbers are sequential across the series; OpsCo is a coordination class and carries no FC number, mirroring DBA-ES-CntlCo. Grounded in the CIKR_v16 sector map (Dams Core Infrastructure and Operational Characteristics).

Section 9 — Development Status and Open Items

This is the initial working draft of the DBA-DAMS sector framework, grounded in the CIKR_v16 sector map, and it completes the corpus's representation of all sixteen NIPP sectors. It establishes the standalone-series placement, the stored-energy catastrophic-release consequence with the Kakhovka reference event, the consolidation of the dam-safety consequence that the electric, water, and transportation series each touch but none owns, the function-primary decomposition (dam structures and water-control works, reservoirs, levees and retention works, and the OpsCo dam-safety-and-emergency-action class) with a hazard-potential sub-index, and the inundation-zone consequence measure. The design-basis envelope (Section 4), the dual-DB detail (Section 5), and the MA applicability section (Section 7) are stubs to be developed. DBA-DAMS-DS-FC1 (*Dam Structures and Water-Control Works*) has issued as the series' first facility-class instrument. It develops the inundation consequence measure of Section 3.6 for the deliberate case and holds the shared model provisionally at its Section 4, and the four reference events it proposed are conferred at Section 9.1. What it notes to this framework is taken up at the framework's next conformance pass: the event categories it develops ahead of Section 4, the model itself, the finding that inundation information serves as a targeting aid, the consequence-basis conditionality check, and its reading of the Section 5 statement on gate openings by the consequence of each command, which this framework is to confirm or replace. Open items at this version: confer a designation on a standalone inundation model instrument; adopt or replace the boundary *Dam Structures and Water-Control Works* proposes with OpsCo, under which the instruments on the structure are the class's hardware and the detection and warning they feed are OpsCo's; agree with Project 8 whether the dam-safety consequence chain of DBA-ES-HY-FC1 moves to this series, is cited here, or stays with a pointer; formalize the boundary protocol with DBA-W-FC3 and DBA-TRAN; and write the four reference-event instruments of Section 9.1.

9.1 Conferred Reference Events

On the author's decision of 12 October 2026, this framework confers a reference-event designator on each of the four events its first facility class proposed. A designator commits the series to two things: a row in the corpus-wide DBA Reference Event Register, and a standalone reference-event instrument for that event. Under the determination of 9 October 2026 (*Reference Events Move to Their Own Project*), reference-event instruments are written and held by Project 11, DBA Reference Events; this framework keeps the conferral. None of the four instruments is yet written. The designators are numbered in the order the proposing classes were issued and, within each class, in the order that class lists its events. The number records conferral, not issue, and the instruments may be written in any order. Each designator is sector-primary and single: an event whose consequence reached other sectors is conferred here, in the sector that proposed it, and its reach is recorded in the register's impacted-sector column rather than in a second designator. That applies to three of the four. Kakhovka destroyed a hydroelectric plant, took the cooling-water source of a nuclear plant and cut off water supply drawn from the reservoir;

Oroville shut down the powerplant at the State Water Project's largest reservoir; and the Edenville and Sanford floods reached a chemical plant's containment ponds at Midland. The impacted sectors are for the instruments to propose and for Project 0 to enter.

From *Dam Structures and Water-Control Works*, four events are conferred: DBA-DAMS-RE-1, the destruction of the Kakhovka dam on the Dnipro on 6 June 2023; DBA-DAMS-RE-2, the Oroville spillway incident in California of February 2017; DBA-DAMS-RE-3, the failures of the Edenville and Sanford dams on the Tittabawassee River in Michigan on 19 May 2020; and DBA-DAMS-RE-4, the intrusion into the control system of the Bowman Avenue Dam in Rye Brook, New York, in August and September 2013. Kakhovka is the event this framework has named since its first version as the sector's reference event (Sections 1.5 and 7); conferral gives it a designator and an instrument. The four instruments are owed, and they are Project 11's to write. *Dam Structures and Water-Control Works* states the four as proposed at its v0.1 and conforms to the conferral at its next revision.

This framework is still a skeleton at Sections 4, 5 and 7. The four instruments read their events back through the design basis of *Dam Structures and Water-Control Works*, which develops it ahead of the framework, and what they return to the framework is taken up at its next conformance pass.

References

[To be completed.] Governing and inherited instruments: FD-BL and determination annexes; DBA-MA Parent Framework (IAEA L1); DBA-ES-HY-FC1 (hydroelectric boundary; Kakhovka precedent already cited); DBA-W-FC3 (water-storage boundary); DBA-TRAN (navigation-locks boundary); DBA-MA nuclear series (Kakhovka–Zaporizhzhia coupling); DBA-COMM-CDB (emergency-notification degradation). Regulatory: FERC (hydro licensing); USACE (dam safety; national inventory); Bureau of Reclamation; state dam-safety programs; Emergency Action Plan requirements; hazard-potential classification. Grounding: CIKR_v16 sector map (Dams Sector). Series instruments: DBA-DAMS-DS-FC1 (*Dam Structures and Water-Control Works*), the series' first facility class and the provisional home of the inundation consequence model (Section 3.6); DBA-DAMS-RE-1 to DBA-DAMS-RE-4, the conferred reference events of Section 9.1, their instruments owed to Project 11.

Revision History

0.4 October 2026 — T. Roxey. Section 9.1 is added and confers DBA-DAMS-RE-1 to DBA-DAMS-RE-4 on the four reference events that DBA-DAMS-DS-FC1 (Dam Structures and Water-Control Works) proposed, on the author's decision of 12 October 2026: the destruction of the Kakhovka dam (June 2023), the Oroville spillway incident (February 2017), the failures of the Edenville and Sanford dams (May 2020) and the intrusion into the Bowman Avenue Dam's control system (2013). Each commits the series to a register row and a standalone instrument, none yet written; the instruments are Project 11's to write under the determination of 9 October 2026, and the conferral stays here. They are numbered in the order the class lists them, and each is conferred in this sector, which proposed it, including the three whose consequence also

reached Energy, Water or Chemical. Section 3.6 names that class's Section 4 as the provisional home of the shared inundation consequence model, on the author's decision of the same date. The register at Section 8.2 enters the class as issued. Section 9 records its issue, lists what it notes to this framework for the next conformance pass, and restates the open items. References add the class and the four designators. The class conforms to the conferral at its next revision. No change to the taxonomy, the designators of the classes, the hazard-potential sub-index or the design basis. Revision-history entries are left as records.

0.3 October 2026 — T. Roxey. Conformed to the October 2026 Edition of the Foundational Definitions on Project 0's work order of 3 October 2026. Section 5 states the scope of Bright Line placement by the determinism carve-out of FD-LD §4.3 rather than as any AI action, and states that determinism decides scope and not placement (P9-11, wording). Functional class is retired at every site under the author's determination of 30 September 2026, each site read for its sense: the decomposition tier reads function class, and the coordination layer reads coordination class (OI-15). No change to the design basis, the taxonomy or the register.

0.2 August 2026 — T. Roxey. Designators ratified; reservoir class code RS → RM; coordination resolved to OpsCo (dam-safety consolidation across owners under FERC/state/FEMA); DS / RM / LV confirmed; stored-energy catastrophic-release framing confirmed.

0.1 August 2026 — T. Roxey. Initial skeleton and the sixteenth NIPP sector, completing the set; grounded in the CIKR_v16 sector map. Established DBA-DAMS as a standalone sector series with the stored-energy catastrophic-release consequence (a peak, mass-casualty inundation event, not a recovery-duration loss; the 2023 Kakhovka breach as the reference event and the dam analog of Zaporizhzhia) and as the consolidation point for the dam-safety consequence that DBA-ES-HY-FC1, DBA-W-FC3, and DBA-TRAN each touch but none owns. Adopted the function-primary decomposition (Dam Structures & Water-Control Works / Reservoirs & Appurtenant Works / Levees, Floodwalls & Retention Works / OpsCo dam-safety-and-emergency-action) with the hazard-potential classification as the sub-index and criticality anchor; adopted the inundation-zone consequence measure; named the electric, water, and transportation boundaries. Design-basis envelope, dual-DB detail, and MA applicability sections stubbed for development.